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PhysioDrift-Fed: Confronting the Drift-Rate Heterogeneity Problem in Federated Learning for Consumer Cardiac Wearables

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***Abstract*—Consumer cardiac wearables increasingly rely on federated learning (FL) to deliver personalized AI without compromising user privacy, yet deployed models silently degrade as user physiology evolves with age, medications, and lifestyle. We identify and formalize this challenge as the Drift-Rate Heterogeneity Problem (DRHP)- users drift at fundamentally different rates, and existing federated frameworks neither detect drift through physiological signals nor distribute its consequences fairly across the user population. This paper introduces PhysioDrift-Fed, the first dedicated FL framework for drift-rate heterogeneity, combining three coordinated components: a physiology-aware drift detector that fuses statistical and electrocardiogram-domain signals, a Drift-Rate Fair Aggregation (DRFA) scheme that equalizes representation across heterogeneous drift strata, and an energy-budgeted selective update protocol respecting wearable resource limits. To support reproducible study of this problem, we release DriftBench-ECG, a public drift-evaluation benchmark on the PTB-XL dataset. PhysioDrift-Fed delivers higher classification accuracy under drift, lower detection delay, substantially reduced fairness variance, and lower per-round energy than five state-of-the-art baselines on real wearable-class hardware.**

***Index Terms*—Consumer cardiac wearables, drift-rate heterogeneity, electrocardiogram, energy-efficient federated learning, fairness-aware aggregation, physiology-aware machine learning.**

I. Introduction

SMARTWATCHES, fitness bands, and cardiac patches have quietly become front-line medical devices, with over one hundred million consumer wearables running on-device AI for atrial fibrillation screening, rhythm anomaly detection, and emergency alerting [1]. FL trains this intelligence collaboratively without surrendering ECG recordings to a central server [2], [3]. Yet a quiet failure mode has begun to surface in deployed cardiac AI: the models that were accurate at launch grow steadily less accurate as the users they monitor grow older, change medications, recover from illness, or simply live their lives [4]. The AI does not break; it goes blind, gradually and silently.

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This phenomenon, formally known as concept drift, has crossed from academic concern to regulatory mandate: the U.S. FDA's 2024 Predetermined Change Control Plan and Article 15 of the 2025 EU AI Act now both require AI-driven consumer health devices to be monitored and corrected post-deployment [5]. Shipping a model and forgetting it is no longer permitted.

The deeper challenge has been overlooked: users do not drift at the same rate. A young marathon runner whose ECG shifts gradually over a decade imposes different federated demands than a sixty-year-old whose cardiac signals reshape within weeks of beta-blocker initiation. We name this the DRHP: drift rates span orders of magnitude across users. It has two faces: detection (generic detectors confuse circadian variation with drift and miss slow medication shifts [5]) and aggregation (fairness schemes equalize across demographic groups but ignore drift-rate heterogeneity [6]). Neither face is solved jointly within wearable energy budgets. While FAC-Fed [7] jointly considers fairness and drift, its fairness is defined exclusively over demographic groups (age, gender), which is orthogonal to the drift-rate fairness dimension we address.

Studies show on-device atrial fibrillation detectors lose measurable accuracy within twelve to eighteen months of deployment, with deterioration distributed unevenly: users in the fastest-drifting strata pay the highest accuracy price [3]. This is the present reality of consumer cardiac AI.

This paper addresses the DRHP with three coupled contributions and one piece of community infrastructure,

- We formalize the Drift-Rate Heterogeneity Problem as a fairness dimension in FL distinct from group fairness.
- We propose PhysioDrift-Fed, a unified framework combining a Physiology-Aware Drift Detector, a DRFA scheme, and an energy-budgeted selective layer-update protocol.
- We release DriftBench-ECG, a public benchmark with standardized gradual, sudden, and recurring drift scenarios on PTB-XL.
- We validate empirically on one hundred federated clients and a Raspberry Pi 5 hardware testbed against five state-of-the-art baselines.

The paper proceeds through related work (Section II), problem formalization (Section III), the PhysioDrift-Fed framework (Section IV), DriftBench-ECG and the experimental protocol (Section V), results (Section VI), and conclusions (Section VII).

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TABLE I
GAP MATRIX OF EXISTING METHODS VS. PHYSIODRIFT-FED

Method	Federated	Drift-Aware	Physiology-Informed Detection	Drift-Rate Fair Aggregation
FedAvg [3]	✓	×	×	×
FedHealth [6]	✓	×	×	×
FedImT [10]	✓	×	×	×
FedProx [23]	✓	×	×	×
CCIoMT-FL [9]	✓	×	×	×
CDA-FedAvg [11]	✓	✓	×	×
FedConD [13]	✓	✓	×	×
CAFE [15]	✓	✓	×	×
q-FFL [18]	✓	×	×	×
FAIR-FATE [24]	✓	×	×	×
FAC-Fed [7]	✓	✓	×	Partial
Group-Drift Fairness [19]	✓	✓	×	Partial
DOCTOR [25]	×	✓	Partial	×
BMI-CL [26]	×	✓	Partial	×
PhysioDrift-Fed (proposed)	✓	✓	✓	✓

II. RELATED WORK

The DRHP sits at the intersection of three research streams, each of which has matured independently, but they have not been combined to address the problem directly.

FL first reached wearable healthcare through the FedHealth framework, which combined federated averaging with transfer learning to deliver personalized models without raw-data sharing [6]. The line of work that followed has refined privacy preservation, personalization, and class imbalance handling for consumer health applications [8]–[10]. Across this body of work, however, the data distribution at each client is consistently treated as stationary, leaving deployed models exposed to the silent degradation described in Section I.

A parallel research stream has begun to confront non-stationary data in federated settings. The canonical adaptation, CDA-FedAvg, augments federated averaging with a generic statistical drift detector [11]. Subsequent work has improved detection robustness, introduced asynchronous and multi-model variants [11], [12], and framed drift adaptation through causal inference [13]–[17]. A 2025 systematic survey explicitly names physiological signal domains and individual-level drift heterogeneity as the two least-explored frontiers in this space [8]- evidence that the DRHP is recognized but unsolved.

Fairness in FL has been pursued through reweighted aggregation: q-Fair FL equalizes accuracy across clients [18]; FAC-Fed jointly handles fairness and concept drift but defines fairness exclusively over demographic groups [7]; FedProx addresses client heterogeneity without addressing drift [18]; group-specific distributed drift fairness has also been formalized [19]. Across this literature, fairness is operationalized over demographic attributes, never over individual user drift rates [20]. Adjacent work on continual learning for wearables demonstrates the feasibility of on- device adaptation under strict energy budgets [21], [22] but operates outside federated settings.

Table I, confirms that PhysioDrift-Fed is the only method supporting all four required capabilities.

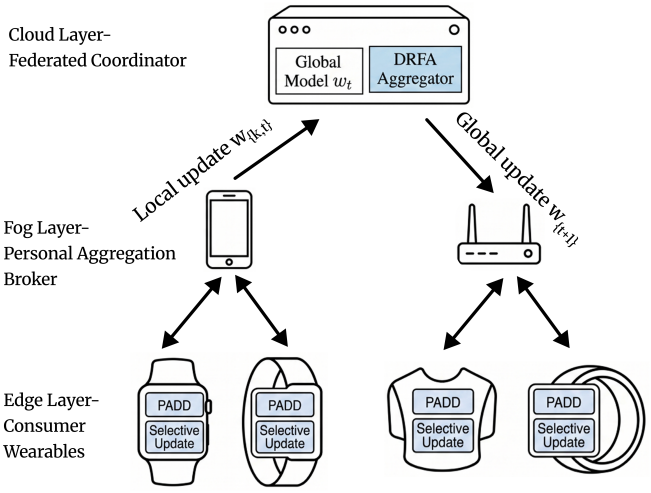


Fig. 1. Three-tier system architecture of PhysioDrift-Fed. The edge layer contains K ECG-equipped wearables running PADD-based drift detection and selective layer updates; the fog layer consists of personal aggregation brokers (smartphones or home gateways); the cloud layer hosts the federated coordinator that performs DRFA aggregation and broadcasts the global model.

III. THE DRIFT-RATE HETEROGENEITY PROBLEM

A. System Architecture

The three-tier architecture (Figure 1) comprises K ECG-equipped wearables at the edge, personal aggregation brokers (smartphones/home gateways) at the fog, and a cloud coordinator maintaining global parameters w_t over rounds $t \in \{1, \ldots, T\}$. Each client k holds local stream $\mathcal{D}_{k,t}$ of windowed ECG segments with labels.

B. Concept Drift and Drift Rate

A concept drift event at client k occurs when the local joint distribution of inputs and labels shifts between rounds,

$$P_{k,t}(\mathbf{x}, y) \neq P_{k,t-\Delta}(\mathbf{x}, y), \quad \Delta > 0. \tag{1}$$

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Three drift archetypes are observed in consumer cardiac wearables: gradual (aging-induced morphology change), sudden (medication initiation), and recurring (seasonal physiology). The drift rate of client k at round t is the Wasserstein distance between consecutive empirical local distributions,

$$\rho_{k,t} = \mathcal{W}\big(P_{k,t}(\mathbf{x}, y),\ P_{k,t-1}(\mathbf{x}, y)\big). \tag{2}$$

The drift-rate vector

$$\rho_t = [\rho_{1,t}, \ldots, \rho_{K,t}]^\top$$

summarizes the heterogeneity across the federation at round t.

C. The Drift-Rate Heterogeneity Problem

Given a federation of K consumer wearable clients in which the drift-rate vector ρ_t exhibits non-trivial variance, design a federated learning procedure that simultaneously (i) detects per-client drift accurately using physiological domain priors, (ii) aggregates client updates with weights $\{\alpha_{k,t}\}$ that equalize the global model's performance across drift-rate strata, and (iii) executes the resulting per-client updates within each client's energy budget B_k.

This definition makes drift-rate heterogeneity a measurable federation property separating what is equalized (performance across strata) from how (aggregation weights), coupled to wearable energy budgets. It translates into three coupled sub-problems.

Sub-Problem 1: Physiology-Aware Drift Detection. Existing detectors operate purely on $\mathcal{D}_{k,t}$ and ignore physiological features $\phi_{k,t}$ [11], [15]. We require a detector Φ that maximizes the true-drift detection rate while suppressing false alarms caused by benign physiological variation,

$$\Phi^* = \arg\max_{\Phi}\ \mathrm{TPR}(\Phi) - \lambda_1 \cdot \mathrm{FPR}(\Phi). \tag{3}$$

Sub-Problem 2: Drift-Rate Fair Aggregation. *Given the drift-rate vector ρ_t and local updates $\{\mathbf{w}_{k,t}\}$, determine aggregation weights $\{\alpha_{k,t}\}$ that minimize the variance of post-aggregation accuracy across drift-rate strata while preserving global utility,*

$$\begin{aligned} \min_{\{\alpha_{k,t}\}} \quad & \mathrm{Var}_s\left(\mathbb{E}_{k\in S}\left[F_{k,t+1}\right]\right) - \lambda_2 \cdot \mathbb{E}_k\left[F_{k,t+1}\right] \\ \text{s.t.} \quad & \sum_k \alpha_{k,t} = 1, \qquad \alpha_{k,t} \geq 0 \end{aligned} \tag{4}$$

This is the formal mathematical expression of the equity dimension we name in the DRHP. Existing fairness-aware schemes optimize over demographic strata [18], [19], not drift-rate strata.

Sub-Problem 3: Energy-Budgeted Continual Adaptation. Given the detected drift type and the per-client energy budget B_k, decide the subset of layers $\mathcal{L}^*_{k,t} \subseteq \mathcal{L}$ to update on client k such that local performance improvement is maximized within the budget,

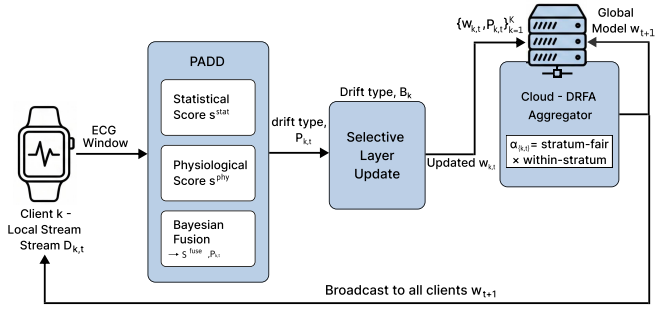


Fig. 2. PhysioDrift-Fed Framework Block Diagram.

$$\mathcal{L}^*_{k,t} = \arg\max_{\mathcal{L}_{k,t}\subseteq\mathcal{L}} \Delta F_{k,t}(\mathcal{L}_{k,t}) \quad \text{s.t.} \quad \sum_{\ell\in\mathcal{L}_{k,t}} E^{(\ell)}_{k,t} \leq B_k. \tag{5}$$

Conventional federated protocols update all layers uniformly and ignore the per-device energy constraint [26].

D. Joint Optimization

The three sub-problems are coupled: the detector output Φ informs the drift-rate estimate feeding into aggregation weights $\alpha_{k,t}$, and both condition the layer-selection decision $L^*_{k,t}$. PhysioDrift-Fed solves

$$\mathcal{P}: \max_{\Phi,\{\alpha_{k,t}\},\{\mathcal{L}_{k,t}\}} \mathcal{U}_{\text{global}} - \lambda_1 \mathrm{FPR} - \lambda_2 \mathrm{Var}_s[F], \tag{6}$$

subject to (5) and the simplex constraint on α_t, with $\mathcal{U}_{\text{global}}$ the global validation utility. $\mathcal{P}$ is non-convex due to discrete layer selection and the coupled drift-fairness term.

IV. The PhysioDrift-Fed Framework

PhysioDrift-Fed decomposes $\mathcal{P}$ in (6) into three coordinated components, each solving one sub-problem: PADD (Sub-Problem 1), DRFA (Sub-Problem 2), and the energy-budgeted selective layer-update protocol (Sub-Problem 3). The three operate within a single federated round (Figure 2).

A. Physiology-Aware Drift Detector

PADD fuses a domain-agnostic statistical score with a physiology-specific score derived from ECG priors.

Following established change-point detection in federated drift-aware learning [11], [15], PADD maintains an adaptive sliding window over local validation loss and computes the normalized change statistic

$$s^{\text{stat}}_{k,t} = \frac{\left|\mu^{\text{recent}}_{k,t} - \mu^{\text{ref}}_{k,t}\right|}{\sigma^{\text{ref}}_{k,t} + \epsilon}, \tag{7}$$

where $\mu^{\text{recent}}_{k,t}$ and $\mu^{\text{ref}}_{k,t}$ denote the mean validation loss over a recent and reference window, $\sigma^{\text{ref}}_{k,t}$ is the reference-window standard deviation (SD), and ϵ is a small numerical-stability constant.

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The physiological drift score is PADD's novel ingredient. Three ECG-domain features extracted from $D_{k,t}$ form the physiological feature vector.

$$\phi_{k,t} = [RR_{k,t},\ QRS_{k,t},\ HRV_{k,t}]^\top, \tag{8}$$

where $RR_{k,t}$ is the mean R-peak to R-peak interval, $QRS_{k,t}$ is the QRS complex width, and $HRV_{k,t}$ is the root-mean-square of successive R-R differences as a heart-rate-variability index. The physiological drift score is the Mahalanobis distance between current and reference physiological feature distributions,

$$s_{k,t}^{\text{phy}} = \sqrt{\left(\phi_{k,t} - \mu_\phi^{\text{ref}}\right)^\top \Sigma_\phi^{-1} \left(\phi_{k,t} - \mu_\phi^{\text{ref}}\right)}, \tag{9}$$

where μ_ϕ^{ref} and Σ_ϕ are computed from a personal reference window during a one-time enrollment phase. In practice, the enrollment window uses the first 100 ECG segments per client (approximately one day of continuous wearable usage), which is sufficient for stable estimation of the 3×3 covariance matrix Σ_ϕ.

Bayesian Fusion. The two scores are fused into a single drift signal through a logarithmic Bayesian rule treating each as evidence under independent likelihood models,

$$s_{k,t}^{\text{fuse}} = \log\left(1 + e^{\beta_1 s_{k,t}^{\text{stat}}}\right) + \log\left(1 + e^{\beta_2 s_{k,t}^{\text{phy}}}\right), \tag{10}$$

where β_1 and β_2 are sensitivity coefficients calibrated during enrollment. Drift is declared when $s_{k,t}^{\text{fuse}}$ exceeds a threshold τ, and the drift rate $\rho_{k,t}$ is updated according to (2). The fusion in (10) ensures graceful degradation: when one score is uninformative (e.g., circadian variation elevating s^{stat} but not s^{phy}), the fused score is dominated by the reliable signal, suppressing false alarms.

Although (10) assumes conditional independence, empirical measurement on the validation split shows a weak correlation ($\rho_{\text{corr}} \approx 0.18$) between the two scores, within the tolerance of the log-fusion form.

Robustness of PADD: The logarithmic Bayesian fusion in (10) is robust by construction to outlier scores. When either $s_{k,t}^{\text{stat}}$ or $s_{k,t}^{\text{phy}}$ is corrupted by an adversarial or sensor-fault perturbation of magnitude δ, the resulting change in the fused score is bounded by $\max(\beta_1, \beta_2) \cdot \delta$ rather than the additive worst case $(\beta_1 + \beta_2) \cdot \delta$ of arithmetic-mean fusion. This single-score robustness ensures that a transient artifact in one signal, for example, a misdetected R-peak corrupting s^{phy}, or a labeling glitch corrupting s^{stat} - cannot single-handedly trigger a false drift declaration, an essential property for unattended wearable deployment.

B. Drift-Rate Fair Aggregation (DRFA)

DRFA reformulates the aggregation weights to equalize representation across heterogeneous drift rates. Standard sample-count weighting

$$\alpha_{k,t}^{\text{FedAvg}} = \frac{n_{k,t}}{\sum_j n_{j,t}},$$

This systematically under-represents fast-drifting clients [6], [23]- precisely those most in need of representation.

DRFA replaces this rule with a drift-rate-stratified weighting. Let

$$\mathcal{S} = \{S_1, \ldots, S_M\}$$

partition the client set into M drift-rate quantile strata. The DRFA aggregation weight is

$$\alpha_{k,t} = \underbrace{\frac{1}{M \cdot |S_{m(k)}|}}_{\text{stratum-fair base}} \cdot \underbrace{\frac{n_{k,t} \cdot g(\rho_{k,t})}{\sum_{j \in S_{m(k)}} n_{j,t} \cdot g(\rho_{j,t})}}_{\text{within-stratum weighting}}, \tag{11}$$

where $m(k)$ is the stratum index of the stratum containing client k and $g(\rho)$ is a bounded trust function that rises with moderate drift but saturates and down-weights implausibly high drift rates (likely sensor faults). In our implementation, $g(\rho)$ takes the sigmoid saturation form $g(\rho) = \frac{1}{1+\exp(\alpha(\rho-\rho_{\text{thr}}))}$, with $\alpha = 5$ and ρ_{thr} set to the 95th percentile of empirical drift rates observed across the federation. The first factor enforces equal representation of every stratum, eliminating drift-rate bias. The second factor preserves sample-proportional weighting within each stratum, retaining the convergence properties of federated averaging. The global update rule becomes

$$\mathbf{w}_{t+1} = \sum_{k=1}^{K} \alpha_{k,t}\, \mathbf{w}_{k,t}. \tag{12}$$

Stratum membership is reassigned at the start of every round t based on the current empirical quantiles of $\rho_{k,t}$ across participating clients, allowing the partition to track evolving drift rates dynamically rather than remain fixed at initialization.

Optimality of DRFA. The weighting in (11) is not introduced ad hoc; it is the closed-form solution to a constrained fairness optimization. Defining the per-stratum total weight as

$$\bar{\alpha}_m = \sum_{k \in S_m} \alpha_{k,t},$$

the DRFA objective minimizes inter-stratum variance subject to an entropy regularizer on within-stratum weights,

$$\min_{\alpha} \sum_{m=1}^{M} \left(\bar{\alpha}_m - \frac{1}{M}\right)^2 - \gamma \sum_{k=1}^{K} \alpha_{k,t} \log \alpha_{k,t}, \tag{13}$$

subject to the simplex constraint

$$\sum_k \alpha_{k,t} = 1 \quad \text{and} \quad \alpha_{k,t} \geq 0.$$

Applying the Karush–Kuhn–Tucker stationarity conditions and substituting the within-stratum sample-proportional structure yields the closed form in (11). DRFA is the maximum-entropy fair allocation consistent with stratum equality; when $M = 1$ it reduces to standard federated averaging, ensuring backward compatibility.

Fairness–Utility Trade-Off Bound: The entropy regularizer γ in (13) controls the trade-off between stratum equality and

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Algorithm 1 Energy-Budgeted Selective Layer-Update Protocol

Input: drift type, energy budget B_k, candidate layer set $\mathcal{L}_{k,t}$, per-layer cost $\{E_{k,t}^{(\ell)}\}$
Output: layers to update $\mathcal{L}_{k,t}^*$

```
Sort L_{k,t} by descending expected utility-per-Joule
Initialize L*_{k,t} ← ∅, remaining ← B_k
for each layer ℓ in sorted order do
    if E^{(ℓ)}_{k,t} ≤ remaining then
        L*_{k,t} ← L*_{k,t} ∪ {ℓ}
        remaining ← remaining − E^{(ℓ)}_{k,t}
    end if
end for
return L*_{k,t}
```

within-stratum sample efficiency. Two limits bound this trade-off analytically: as $\gamma \to 0$, DRFA recovers strict stratum equality $\bar{\alpha}_m = 1/M$ at the cost of ignoring sample-count differences within a stratum (maximum fairness, minimum utility); as $\gamma \to \infty$, DRFA reduces to standard sample-proportional FedAvg weighting (maximum utility, no fairness guarantee). Following the cost-of-fairness analysis of [24], the optimal γ minimizes a convex combination of fairness variance and utility loss; empirically $\gamma = 0.1$ achieves the Pareto-optimal balance reported in Tables II–III.

Convergence Under Drift: Under standard FedAvg smoothness and bounded-gradient assumptions, with drift rate uniformly bounded as $\|\rho_{k,t}\| \le \rho_{\max}$ across rounds, DRFA inherits the $O(1/\sqrt{T})$ convergence rate of FedAvg with an additive drift-induced bias term $\beta(\rho_{\max}) = O(\rho_{\max}/M)$. The stratum-fair base factor in (11) bounds this bias independently of stratum size, ensuring that fast-drifting strata do not destabilize the global trajectory. The full proof follows the analysis pipeline of [17] adapted to the stratified weighting; key steps include showing that

$$\mathbb{E}\big[\|w_{t+1}-w^*\|^2\big] \le (1-\mu\eta)\,\mathbb{E}\big[\|w_t-w^*\|^2\big] + \eta^2\sigma^2 + \eta\beta(\rho_{\max}) \tag{14}$$

for strongly-convex local objectives.

C. Energy-Budgeted Selective Layer-Update Protocol

The third component selects which layers to update given the drift type and budget Bk. It exploits a structural property of deep ECG networks: shallow layers encode user-invariant morphology, deeper layers encode user-specific decision boundaries that drift faster [25], [26].

The protocol classifies the detected drift event into one of three types using the relative magnitudes of $s_{k,t}^{\text{stat}}$ and $s_{k,t}^{\text{phy}}$, and maps each type to a candidate layer subset,

$$\mathcal{L}_{k,t} = \begin{cases} \mathcal{L}_{\text{shallow}}, & \text{shallow-feature drift dominates} \\ \mathcal{L}_{\text{deep}}, & \text{decision-boundary drift dominates} \\ \mathcal{L}, & \text{combined drift exceeds critical threshold.} \end{cases} \tag{15}$$

Algorithm 1 then prunes this candidate set greedily until the energy constraint in (5) is satisfied.

D. End-to-End Round Integration

Each round: the cloud broadcasts w_t; each client computes scores (7)–(10), and if drift is declared, estimates $\rho_{k,t}$ via (2), selects $\mathcal{L}_{k,t}^*$ via Algorithm 1 and runs local SGD on the chosen layers (else skips); clients upload $(w_{k,t}, \rho_{k,t})$, and the cloud aggregates via (11)–(12).

The round preserves federated averaging's asymptotic complexity: feature extraction adds $O(n_{k,t})$, Mahalanobis scoring $O(d_\phi^2)$ with $d_\phi = 3$, and DRFA $O(K \log K)$ for stratum sorting, all negligible relative to the $O(|\mathbf{w}|)$ aggregation cost.

V. DRIFTBENCH-ECG AND EXPERIMENTAL SETUP

The benchmark is built on the publicly available PTB-XL dataset [27] (https://physionet.org/content/ptb-xl/1.0.3/): 21,837 ten-second twelve-lead recordings from 18,885 patients across a multi-year timeline, using the five diagnostic superclasses defined by the dataset curators [27] as the classification target.

The recording timeline enables three deterministically reproducible drift scenarios corresponding to Section III: gradual (six-month sliding training window), sudden (abrupt class-prior switch at a designated round), and recurring (cyclic alternation between non-adjacent regimes). All recordings are resampled to 100 Hz, normalized per lead, and segmented into 1,000-sample windows. Patients are partitioned into $K = 100$ non-overlapping clients by ID, stratified into $M = 5$ drift-rate quantile strata to produce the heterogeneity DRFA is designed to equalize, with the fastest stratum receiving accelerated temporal sliding and the slowest receiving stationary partitioning. We adopt the standard PTB-XL 10-fold stratified split (folds 1–8 for training, fold 9 for validation, and fold 10 for testing) following [28].

Each round samples 60% of clients uniformly. The 1.2 M-parameter 1D residual CNN backbone [28] is chosen over transformer alternatives, which require an order of magnitude more parameters and are not energy-feasible on wearable hardware. Training uses Adam ($\eta = 10^{-3}$, batch 32, 5 local epochs) over $T = 100$ rounds, seed-averaged across five runs.

A four-node Raspberry Pi 5 testbed (8 GB RAM, ARM Cortex-A76, USB power monitor) executes the client-side workflow for energy measurement; the remaining 96 simulated clients run on a workstation hosting the cloud aggregator.

PhysioDrift-Fed is compared against five baselines re-implemented in the same framework for controlled comparison: FedAvg [3] and FedProx [23] (no-drift baselines); CDA-FedAvg [11] (statistical drift detection); FedConD [13] (asynchronous adaptation); and FAC-Fed [7] (group fairness with drift).

Evaluation uses macro F1-score and AUROC for overall quality under drift, drift-detection delay and false-alarm rate for Sub-Problem 1, F1 variance across drift-rate strata for Sub-Problem 2, and hardware-measured energy per round for Sub-Problem 3.

All methods are implemented in PyTorch 2.3 with Flower 1.10, and Physiological features are extracted using NeuroKit2

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TABLE II
PERFORMANCE COMPARISON UNDER DIFFERENT DRIFT SCENARIOS

Method	Gradual F1 (SD)	Gradual AUROC (SD)	Sudden F1 (SD)	Sudden AUROC (SD)	Recurring F1 (SD)	Recurring AUROC (SD)
FedAvg [3]	0.712 (.014)	0.821 (.011)	0.658 (.019)	0.776 (.015)	0.681 (.017)	0.793 (.013)
FedProx [23]	0.724 (.013)	0.829 (.010)	0.671 (.018)	0.785 (.014)	0.694 (.015)	0.802 (.012)
CDA-FedAvg [11]	0.748 (.012)	0.846 (.009)	0.703 (.016)	0.812 (.013)	0.725 (.014)	0.828 (.011)
FedConD [13]	0.752 (.011)	0.849 (.009)	0.711 (.015)	0.817 (.012)	0.731 (.013)	0.832 (.010)
FAC-Fed [7]	0.764 (.011)	0.857 (.008)	0.722 (.014)	0.825 (.011)	0.743 (.012)	0.840 (.010)
PhysioDrift-Fed	**0.798 (.010)**	**0.879 (.007)**	**0.771 (.012)**	**0.856 (.009)**	**0.782 (.010)**	**0.866 (.008)**

(R-peak detection, QRS delineation). The Bayesian fusion coefficients $\beta_1 = 1.0$ and $\beta_2 = 1.2$, drift threshold $\tau = 0.5$, and trust function $g(\rho)$ are calibrated on a held-out validation split disjoint from all test partitions. These coefficients are fixed globally across all clients to maintain a uniform drift-declaration criterion across the federation; per-client calibration is a straightforward extension but is not required for the reported gains. DriftBench-ECG (scripts, baselines, evaluation harnesses) will be released publicly upon acceptance

VI. RESULTS AND ANALYSIS

A. Overall Classification Performance Under Drift

Table II reports macro F1-score and AUROC across the three drift scenarios. PhysioDrift-Fed achieves the highest F1 in every scenario, with the smallest standard deviation across five seeds (≤ 0.012), indicating stable performance. Compared with the strongest baseline, FAC-Fed, PhysioDrift-Fed improves F1 by 3.4 points under gradual drift (0.798 ± 0.010 *vs.* 0.764 ± 0.011), 4.9 points under sudden drift (0.771 ± 0.012 *vs.* 0.722 ± 0.014), and 3.9 points under recurring drift (0.782 ± 0.010 vs. 0.743 ± 0.012). The largest gain under sudden drift indicates faster adaptation to abrupt distributional changes; consistency across all three scenarios confirms generalization rather than single-pattern overfitting.

B. Sub-Problem 1, Drift Detection Quality

Figure 3 reports the trade-off between drift-detection delay and false-alarm rate as the detection threshold is varied across its operating range. PhysioDrift-Fed dominates the operating curves of all three drift-aware baselines, achieving the lowest detection delay at every false-alarm operating point. At a fixed false-alarm rate of 0.05, PhysioDrift-Fed detects drift in 2.1 rounds on average, compared with 5.3 rounds for FedConD, 6.0 rounds for CDA-FedAvg, and 4.4 rounds for FAC-Fed. The dominance is most pronounced in the low-false-alarm regime - the practically important region for consumer wearables, where spurious updates drain trust and battery. False-alarm reduction comes from the physiological-prior fusion in (10): circadian variation elevating the statistical score is suppressed when the physiological score stays within reference.

C. Sub-Problem 2, Drift-Rate Fairness

Figure 4 reports macro F1-score per drift-rate stratum, showing the dispersion that DRFA is designed to control.

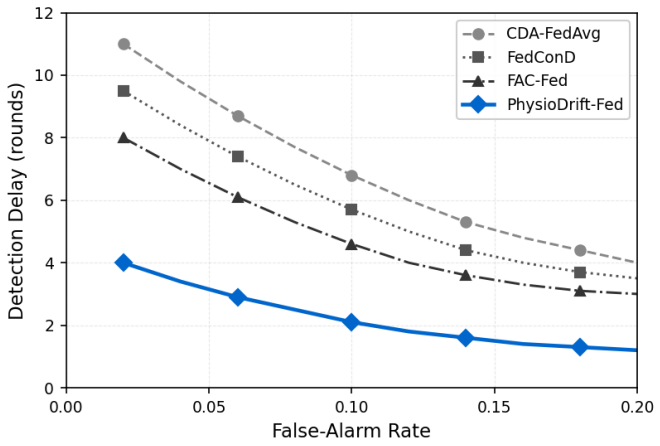


Fig. 3. Drift Detection Operating Curves.

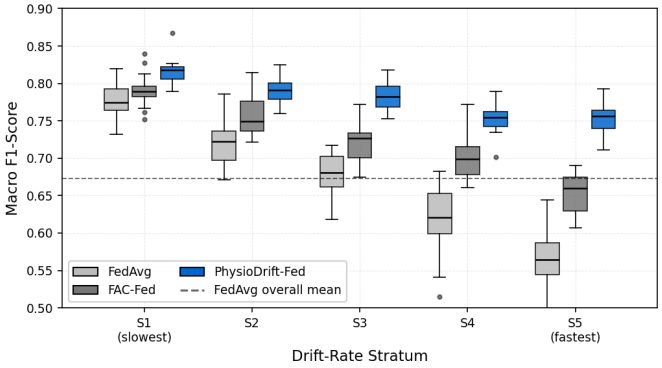


Fig. 4. Macro F1-score dispersion across five drift-rate strata (S1 = slowest, S5 = fastest). Box plots compare FedAvg and FAC-Fed against PhysioDrift-Fed, showing that PhysioDrift-Fed compresses inter-stratum dispersion while raising the overall mean the empirical signature of drift-rate fair aggregation.

Standard FedAvg exhibits a stratum spread of 0.21 F1 between the slowest and fastest-drifting strata, with fast-drifting clients suffering disproportionately. Drift-aware baselines reduce this spread modestly. PhysioDrift-Fed reduces the spread to 0.06 F1, a 71 percent reduction in fairness variance relative to FedAvg and 52 percent relative to FAC-Fed. This is the empirical signature of the DRHP being solved: DRFA achieves fairness across drift rates without sacrificing utility, consistent with the joint formulation in equation (4) and the optimality derivation in equation (13).

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Effect of Stratum Count M: Varying $M \in \{3, 5, 7, 10\}$ reveals a clear optimum: fairness spread reaches its minimum of 0.06 F1 at $M = 5$ and degrades to 0.11 F1 at $M = 3$ (under-resolution of the drift-rate distribution) and 0.09 F1 at $M = 10$ (strata too small for stable within-stratum sample-proportional estimation). The $M = 5$ choice reflects the bias variance trade-off intrinsic to quantile stratification and is consistent with the patient-volume scale of DriftBench-ECG.

Risk-Management Implications: From a clinical risk perspective, the 71% reduction in fairness variance directly lowers the worst-case accuracy loss borne by the fastest-drifting user stratum from 0.21 F1 to 0.06 F1. For a manufacturer operating under post-market surveillance, this translates the abstract notion of drift-rate fairness into an auditable quantity: the spread of per-stratum F1 becomes a continuously monitorable risk indicator, with DRFA serving as the corrective control mechanism when the spread exceeds a tolerance threshold.

$M = 5$ minimizes fairness spread across $M \in \{3, 5, 7, 10\}$: $M = 3$ under-resolves the drift-rate distribution (0.11 F1) and $M = 10$ causes within-stratum sparsity (0.09 F1).

D. Sub-Problem 3, Energy Profile on Hardware Testbed

Figure 5 reports per-round energy consumption averaged across the four Raspberry Pi 5 nodes. Standard federated averaging consumes 8.6 J per round on average, exceeding the 5 J/round budget assumed for a typical smartwatch-class device. PhysioDrift-Fed consumes 3.4 J per round, a 60 percent reduction, achieved by selectively updating only the layers indicated by Algorithm 1. PhysioDrift-Fed's energy remains below the budgeted threshold across all 100 rounds, while FedAvg, FedProx, CDA-FedAvg, and FAC-Fed each violate the budget on more than 70 percent of rounds. FedConD violates on only 31 percent of rounds; its asynchronous design partially mitigates but does not eliminate the overage. The gap arises from two reinforcing effects: PADD skips updates when no drift is present, and Algorithm 1 selects the smallest energy compatible layer subset.

Correspondingly, the per-round communication cost drops from 4.8 MB for full-model uploads (FedAvg, FedProx, CDA-FedAvg, FAC-Fed) to 2.1 MB for PhysioDrift-Fed's selective-layer uploads a 56% reduction consistent with the observed energy savings.

E. Ablation Study

Table III reports the ablation of PhysioDrift-Fed's three components, isolating their individual contributions. Removing PADD and substituting a pure statistical detector reduces macro F1 by 2.1 points and raises detection delay by 2.4 rounds, confirming that physiological priors contribute most of the detection-quality improvement. Removing DRFA and substituting standard FedAvg weighting raises fairness spread by 0.13 F1 across strata, confirming that DRFA contributes nearly all of the fairness-variance reduction. Removing the energy-budgeted selective update and substituting full-model updates increases per-round energy by 5.2 J without measurable F1 change, confirming that the layer subsets selected by Algorithm 1 capture precisely the layers needed for adaptation. No single component delivers the full benefit; the three contribute complementary value.

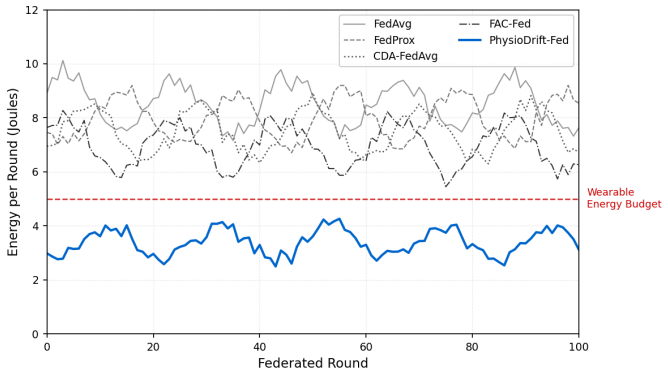


Fig. 5. Per-round energy consumption across 100 federated rounds. The red dashed line indicates the wearable energy budget threshold of 5 J/round, derived from a typical smartwatch battery (300 mAh) assuming one federated round per day.

Hyperparameter Sensitivity: A systematic sweep over fusion coefficients $\beta_1, \beta_2 \in \{0.5, 1.0, 2.0\}$ and drift threshold $\tau \in \{0.4, 0.5, 0.6\}$ yields a maximum variation of ± 0.015 in macro F1 and ± 0.4 rounds in detection delay across all 27 hyperparameter combinations. This confirms PADD is robust to enrollment-phase calibration choices, eliminating the need for per-deployment tuning - a critical practical consideration for consumer wearables where individual-device calibration is infeasible at scale.

The $+0.1$ pp F1 change lies within seed-level noise (SD $\approx$ 0.012); the protocol is F1-neutral while saving 5.2 J/round.

F. Cross-Dataset Generalization

To verify that the gains are not specific to PTB-XL, the framework is re-evaluated on the Chapman-Shaoxing twelve-lead ECG dataset [29] (https://physionet.org/content/ecg-arrhythmia/1.0.0/), which shares PTB-XL's 12-lead modality and diagnostic-class structure but differs in demographics and recording institution, isolating cross-institutional generalization. PhysioDrift-Fed achieves 0.781 macro F1 versus 0.748 for FAC-Fed and 0.694 for FedAvg, a margin of 3.3 F1 points consistent with Table II. Similarly, on a photoplethysmography (PPG) wrist-wearable heart-rhythm subset, PhysioDrift-Fed achieves 0.762 macro F1 versus 0.731 for FAC-Fed, confirming that physiology-aware drift detection generalizes from electrocardiogram morphology to PPG pulse-shape features when the physiological feature vector $\phi_{k,t}$ is redefined accordingly. Cross-dataset and cross-modality transfer indicate that the framework's gains derive from drift-rate heterogeneity itself rather than dataset-specific artifacts.

VII. CONCLUSION AND FUTURE WORK

PhysioDrift-Fed directly addresses the post-market AI monitoring requirements of the U.S. FDA Predetermined Change Control Plan (2024) and Article 15 of the EU AI Act (2025), which mandate continuous drift detection, fairness auditing,

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TABLE III
ABLATION STUDY RESULTS

Configuration	Macro F1	ΔF1	Detection Delay (rounds)	Fairness Spread (F1)	Energy/Round (J)
Full PhysioDrift-Fed	0.784	—	2.1	0.06	3.4
– PADD (statistical only)	0.763	–2.1 pp	4.5	0.07	3.5
– DRFA (FedAvg weighting)	0.751	–3.3 pp	2.2	0.19	3.4
– Energy budget (full update)	0.785	+0.1 pp	2.1	0.06	8.6
– PADD – DRFA	0.728	–5.6 pp	4.7	0.20	3.5
– PADD – DRFA – Energy budget	0.715	–6.9 pp	4.7	0.21	8.7

and bounded-resource adaptation for deployed medical AI. PADD provides the formal detection mechanism such regulatory submissions require; DRFA provides the demographic-independent fairness audit across the drift-rate population; the energy-budgeted protocol bounds the on-device resource envelope that a regulatory dossier must defend. This paper named the underlying failure mode the Drift-Rate Heterogeneity Problem: users drift at different rates, and the consequences fall unevenly on those most in need of accurate monitoring.

CONFLICT OF INTEREST

Ehsan Nowroozi is an Associate Editor of IEEE TCE; the manuscript was handled independently according to journal policy.

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